

Geometry & mensuration

LENGTHS AREAS VOLUMES

A: area P/C: perimeter/circumference V: volume S: total surface area L: lateral/curved area (arc length in circle section) B, p: base area, perimeter

Plane figures



Square
 $A = a^2$, $P = 4a$, $d = a\sqrt{2}$

Rectangle
 $A = lw$, $P = 2(l + w)$
 $d = \sqrt{l^2 + w^2}$

Parallelogram
 $A = bh = ab \sin \theta$, $P = 2(a + b)$
 θ is the included angle of sides a, b .

Trapezium / trapezoid
 $A = \frac{1}{2}(a + b)h$, $P = a + b + c + d$
 a, b are parallel; h is their separation.

Rhombus
 $A = \frac{1}{2}d_1d_2 = ah$, $P = 4a$

Kite
 $A = \frac{1}{2}d_1d_2$, $P = 2(a + b)$
 a, b are the two distinct side lengths.

Regular n-gon
 $P = na$, $A = \frac{1}{2}Pq = \frac{na^2}{4 \tan(\pi/n)}$
 q is the apothem; a is one side.

Ellipse
 $A = \pi ab$
 a, b are semiaxes. No elementary perimeter formula.

Triangle area & centres

Any triangle
 $A = \frac{1}{2}bh = \frac{1}{2}ab \sin C$
 $s = \frac{1}{2}(a + b + c)$
 $A = \sqrt{s(s-a)(s-b)(s-c)}$
Sides a, b, c face angles A_0, B_0, C ; C is between a, b .

Inradius & circumradius
 $A = r_{\text{in}}s = \frac{abc}{4R}$

Equilateral triangle
 $h = \frac{\sqrt{3}}{2}a$, $A = \frac{\sqrt{3}}{4}a^2$
 $r_{\text{in}} = \frac{\sqrt{3}}{6}a$, $R = \frac{\sqrt{3}}{3}a$

Circles, arcs & sectors



Circle & annulus
 $d = 2r$, $C = 2\pi r$, $A = \pi r^2$
 $A_{\text{annulus}} = \pi(R^2 - r^2)$ ($R \geq r$)

Sector - radians
 $L = r\theta$, $A = \frac{1}{2}r^2\theta = \frac{1}{2}rL$
 $P_{\text{sector}} = 2r + L$ ($0 < \theta < 2\pi$)

Sector - degrees
 $L = \frac{\theta_{\text{deg}}}{360} 2\pi r$
 $A = \frac{\theta_{\text{deg}}}{360} \pi r^2$

Chord & minor segment
 $c = 2r \sin(\theta/2)$
 $A_{\text{segment}} = \frac{1}{2}r^2(\theta - \sin \theta)$
Use radians, $0 \leq \theta \leq \pi$; segment = sector minus triangle.

Semicircle
 $A = \frac{1}{2}\pi r^2$, $P = \pi r + 2r$

Angles & trigonometry

Angle conversion
 $180^\circ = \pi \text{ rad}$
 $\theta_{\text{rad}} = \theta_{\text{deg}}\pi/180$

Right triangle
 $c^2 = a^2 + b^2$
 $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$
 $\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin \theta}{\cos \theta}$

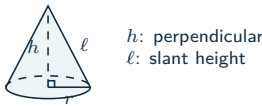
Sine & cosine rules
 $\frac{a}{\sin A_0} = \frac{b}{\sin B_0} = \frac{c}{\sin C} = 2R$
 $c^2 = a^2 + b^2 - 2ab \cos C$

Special right triangles
 $45^\circ : 1 : 1 : \sqrt{2}$
 $30^\circ, 60^\circ, 90^\circ : 1 : \sqrt{3} : 2$
Side ratios in increasing angle order.

Polygon angles
Interior sum $= (n - 2)180^\circ$
Regular interior $= 180^\circ - 360^\circ/n$
Regular exterior $= 360^\circ/n$

Circle angle facts
Central angle $= 2 \times$ inscribed angle
Same arc. An angle in a semicircle is 90° . Opposite angles of a cyclic quadrilateral sum to 180° .

Prisms, cylinders & cones



Cube
 $V = a^3$, $S = 6a^2$, $L = 4a^2$
Space diagonal $= a\sqrt{3}$
Cuboid / rectangular prism
 $V = lwh$, $S = 2(lw + lh + wh)$
 $L = 2h(l + w)$, $d = \sqrt{l^2 + w^2 + h^2}$
Right prism
 $V = Bh$, $L = ph$, $S = ph + 2B$

Right circular cylinder
 $V = \pi r^2h$, $L = 2\pi rh$
 $S = 2\pi r(h + r)$
Open at one end: $S = 2\pi rh + \pi r^2$.

Right circular cone
 $\ell = \sqrt{r^2 + h^2}$
 $V = \frac{1}{3}\pi r^2h$, $L = \pi r\ell$
 $S = \pi r(\ell + r)$

Pyramid
 $V = \frac{1}{3}Bh$
Regular pyramid: $L = \frac{1}{2}pl$, $S = B + L$
 ℓ is a face slant height, not a sloping edge. Otherwise add the areas of all side faces.

Right conical frustum
 $\ell = \sqrt{h^2 + (R - r)^2}$
 $V = \frac{\pi h}{3}(R^2 + Rr + r^2)$
 $L = \pi(R + r)\ell$
 $S = L + \pi R^2 + \pi r^2$

Similar-base pyramidal frustum
 $V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1B_2})$
Parallel bases; h is their perpendicular separation.

Surface & volume checks

Perpendicular height
Use h for volume. Use ℓ for the curved area of a cone or the side faces of a regular pyramid.

Composite shapes
Add or subtract component areas and volumes. For exposed surface area, remove joined faces and include exposed cavity walls.

Spheres & curved solids



Sphere
 $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$
Hemisphere
 $V = \frac{2}{3}\pi r^3$, $L = 2\pi r^2$
 $S = 3\pi r^2$ (including flat base)

Spherical cap
 $V = \frac{\pi h^2}{3}(3R - h)$, $L = 2\pi Rh$
 $a^2 = 2Rh - h^2$, $S = L + \pi a^2$
Sphere radius R ; cap height h ; base radius a ; $0 \leq h \leq 2R$.

Ring torus
 $V = 2\pi^2 Rr^2$, $S = 4\pi^2 Rr$
 $R > r$: R is centre-to-tube-centre radius; r is tube radius.

Coordinate geometry

Distance & midpoint
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Straight lines
 $m = \frac{y_2 - y_1}{x_2 - x_1}$, $y - y_1 = m(x - x_1)$
Parallel: $m_1 = m_2$; $\perp$: $m_1m_2 = -1$
Slope formulas assume finite slopes; vertical lines are perpendicular to horizontal lines.

Circle & point-to-line distance
 $(x - u)^2 + (y - v)^2 = r^2$
 $d = \frac{|\alpha x_0 + \beta y_0 + \gamma|}{\sqrt{\alpha^2 + \beta^2}}$
Point (x_0, y_0) ; line $\alpha x + \beta y + \gamma = 0$; $(\alpha, \beta) \neq (0, 0)$.

Triangle from three vertices
 $A = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Similarity & units

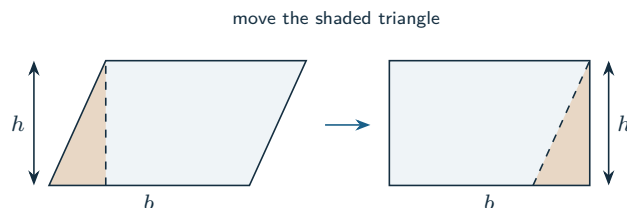
Similar figures / solids
Length k , area k^2 , volume k^3
 k is new length / original length.

Convert before substituting
 $1 \text{ m}^2 = 10^4 \text{ cm}^2$
 $1 \text{ m}^3 = 10^6 \text{ cm}^3 = 10^3 \text{ L}$
 $1 \text{ cm}^3 = 1 \text{ mL}$
Lengths: units; areas: units²; volumes: units³. Keep π exact until the final rounding.

Area comes from rearranging

Cut, duplicate, or subtract a familiar shape. The perpendicular height does the work.

01 A slant does not change the area



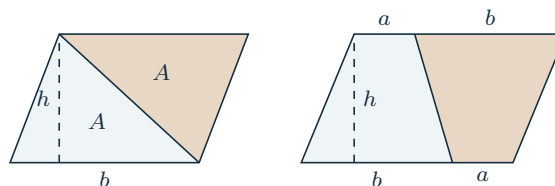
$$A_{\text{parallelogram}} = bh = ab \sin \theta$$

$$A_{\text{rectangle}} = lw, \quad A_{\text{square}} = a^2$$

Cut off one end and slide it to the other. The result is a rectangle with the same base and height. If the sloping side is a , then $h = a \sin \theta$.

See it: A base of 6 and height of 3 give area 18, however far the top slides sideways.

02 Two copies explain the half



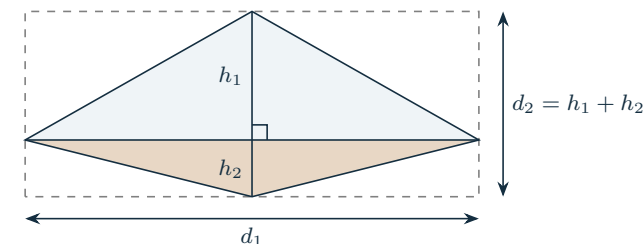
$$A_{\text{triangle}} = \frac{1}{2}bh$$

$$A_{\text{trapezium}} = \frac{1}{2}(a+b)h$$

Two equal triangles make a parallelogram. Two equal trapezia make one with base $a+b$. Divide the doubled area by two.

See it: Parallel sides 3 and 7, height 4: the average width is 5, so the area is $5 \times 4 = 20$.

03 Diagonals split the job

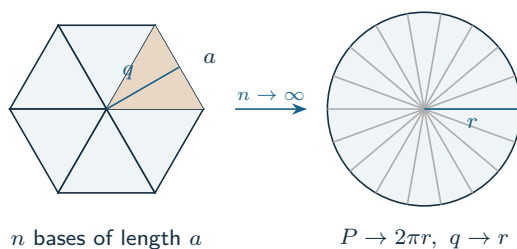


$$A = \frac{1}{2}d_1h_1 + \frac{1}{2}d_1h_2 = \frac{1}{2}d_1d_2$$

The two triangles have the same base d_1 . Their heights add to d_2 . This works for a rhombus or kite because its diagonals are perpendicular.

See it: Diagonals 8 and 5 enclose area 20. A sloping diagonal is not automatically a perpendicular height.

04 Polygons are a fan of triangles



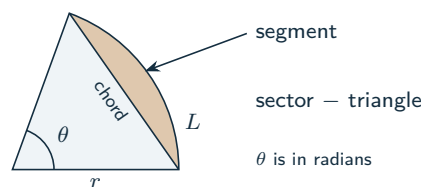
$$A_{n\text{-gon}} = n\left(\frac{1}{2}aq\right) = \frac{1}{2}Pq$$

$$q = \frac{a}{2 \tan(\pi/n)}, \quad A_{\text{circle}} = \pi r^2$$

Each small triangle has altitude q , the apothem. As the number of sides grows, the fan approaches a circle: half the circumference times the radius.

See it: Perimeter $P = 24$, apothem $q = 3$: $A = 36$.

05 A sector is a fraction of a circle



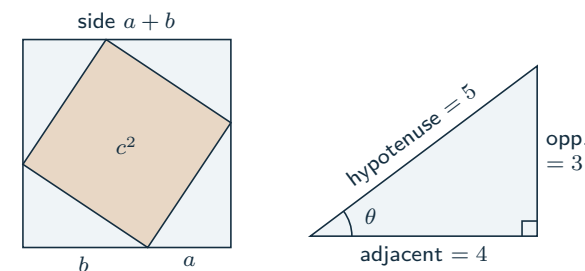
$$L = \frac{\theta}{2\pi}(2\pi r) = r\theta, \quad A_{\text{sector}} = \frac{1}{2}r^2\theta$$

$$A_{\text{segment}} = \frac{1}{2}r^2(\theta - \sin \theta)$$

For the minor segment, subtract the isosceles triangle of area $\frac{1}{2}r^2 \sin \theta$ from the sector ($0 \leq \theta \leq \pi$). A sector's boundary also includes both radii.

See it: At $\theta = \pi/2$: arc $= \pi r/2$; sector area $= \pi r^2/4$.

06 Squares explain lengths; ratios explain angles



$$(a+b)^2 = 4\left(\frac{1}{2}ab\right) + c^2 \Rightarrow a^2 + b^2 = c^2$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

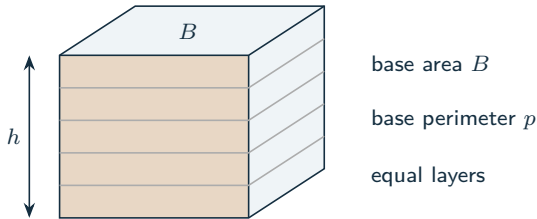
The outer square contains four right triangles and the central square. Similar right triangles keep the same side ratios for the same angle.

See it: For the triangle shown: $\sin \theta = 3/5$, $\cos \theta = 4/5$, $\tan \theta = 3/4$.

Unfold surfaces. Stack volumes.

Surface area counts the skin. Volume counts the space inside.

07 A prism is a stack of equal bases



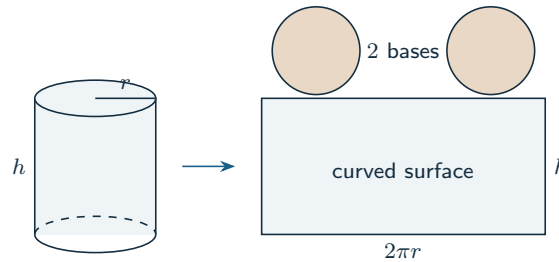
$$V = Bh, \quad L = ph, \quad S = ph + 2B$$

$$V_{\text{cuboid}} = lwh, \quad S_{\text{cuboid}} = 2(lw + lh + wh)$$

Every layer has area B . The side faces unfold into strips whose widths add to p . Add the two bases for a closed right prism.

See it: A cube has six square faces: $S = 6a^2$. Stacking a layers gives $V = a^3$.

08 A cylinder unwraps into a rectangle



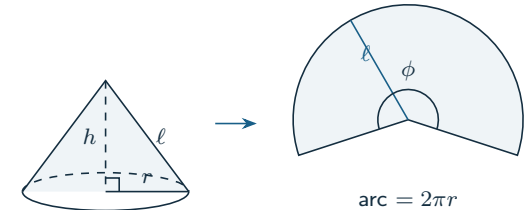
$$L = (2\pi r)h, \quad S = 2\pi rh + 2\pi r^2$$

$$V = \pi r^2 h$$

The rectangle's width is the circumference, not the diameter. A cylinder with one open end has only one circular base to add.

See it: For $r = 2$, $h = 3$: $V = 12\pi$ and closed $S = 20\pi$.

09 A cone's skin is a circular sector



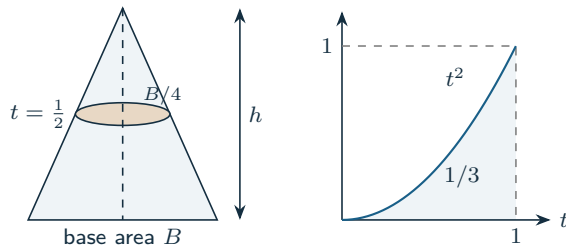
$$\ell = \sqrt{r^2 + h^2}, \quad L = \frac{1}{2}\ell(2\pi r) = \pi r\ell$$

$$\phi = \frac{2\pi r}{\ell}, \quad S = \pi r\ell + \pi r^2$$

The sector radius is the slant height ℓ ; its arc wraps around the base. The perpendicular height h stays inside the cone. The angle ϕ is in radians.

See it: For $r = 3$, $h = 4$: $\ell = 5$, $\phi = 216^\circ$, and $L = 15\pi$.

10 Why cones and pyramids have a third



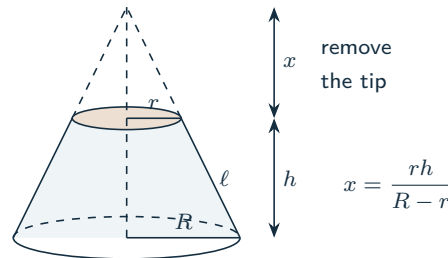
$$A_{\text{slice}} = Bt^2$$

$$V = h \int_0^1 Bt^2 dt = \frac{1}{3}Bh$$

At fraction t of the distance from tip to base, lengths scale by t and slice areas by t^2 . Adding the slices gives one third of the prism with the same B, h .

See it: Halfway from the tip, the slice has one quarter of the base area. The graph shows the slice-area fraction.

11 A frustum is a large cone minus a small one



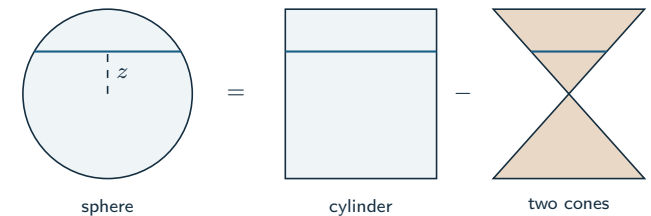
$$V = \frac{\pi}{3} [R^2(h+x) - r^2x]$$

$$= \frac{\pi h}{3} (R^2 + Rr + r^2)$$

$$L = \pi(R+r)\ell, \quad \ell = \sqrt{h^2 + (R-r)^2}$$

Similarity fixes the missing tip's height. Subtract its volume; subtract the two cone skins for L . Add both circular ends for total surface area. Here $R > r$.

12 A sphere can be measured by matching slices



$$A_{\text{slice}} = \pi(R^2 - z^2) = \pi R^2 - \pi z^2$$

$$V = 2\pi R^3 - 2(\frac{1}{3}\pi R^3) = \frac{4}{3}\pi R^3$$

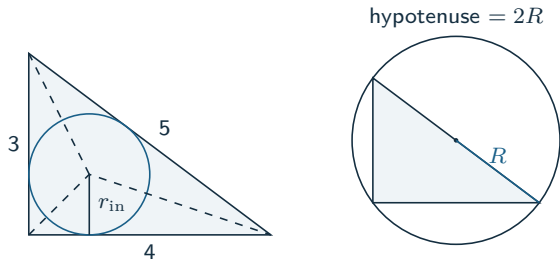
At every height z , the sphere's disk equals the cylinder's disk minus the cone's disk. Matching all slices matches volumes. Each cone has radius and height R .

See it: A sphere fills 2/3 of its enclosing cylinder of radius R and height $2R$. Drawings show vertical sections.

Connect the formulas

Centres, curved surfaces, coordinates, and scale turn separate facts into a small set of ideas.

13 Triangle centres encode the same area



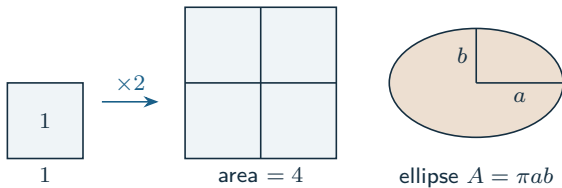
$$A = \frac{1}{2}r_{\text{in}}(a + b + c) = r_{\text{in}}s$$

$$A = \frac{1}{2}bc \sin A_0 = \frac{abc}{4R} \quad (a = 2R \sin A_0)$$

Join the incentre to the vertices: all three small triangles have height r_{in} . The sine rule connects the circumradius to the same area.

See it: For sides 3, 4, 5: $s = 6$, $A = 6$, $r_{\text{in}} = 1$, $R = 2.5$. Heron checks it: $\sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6$.

16 Scaling multiplies once per dimension

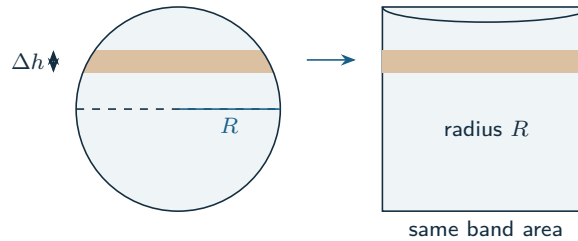


Length: k , area: k^2 , volume: k^3
 $1 \text{ m}^2 = 100^2 \text{ cm}^2$, $1 \text{ m}^3 = 100^3 \text{ cm}^3$

Double every edge: four times the area and eight times the volume. Stretch a unit circle by a horizontally and b vertically: its area becomes πab .

See it: An area ratio of 9 means a length ratio of 3 and a volume ratio of 27 for similar solids.

14 Sphere bands match cylinder bands



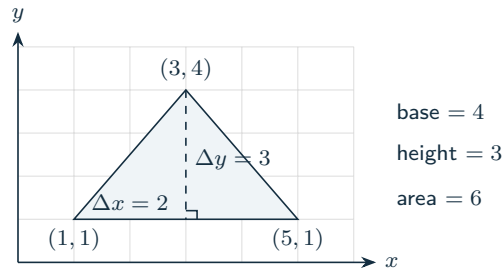
$$\Delta S \approx 2\pi R \Delta h, \quad L_{\text{cap}} = 2\pi R h$$

$$S_{\text{sphere}} = 4\pi R^2, \quad S_{\text{hemisphere}} = 3\pi R^2$$

A sphere band has a smaller circumference but a wider sloping surface. These effects cancel in the thin-band limit. Summing bands gives the curved cap area; h is its vertical height.

See it: Full sphere: $h = 2R$. Hemisphere: curved area $2\pi R^2$ plus the flat base πR^2 .

17 Coordinates turn geometry into differences



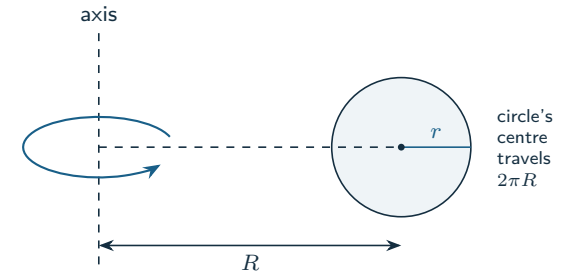
$$d = \sqrt{(\Delta x)^2 + (\Delta y)^2}, \quad m = \Delta y / \Delta x$$

$$A = \frac{1}{2} |\sum x_i y_{i+1} - \sum y_i x_{i+1}|$$

Horizontal and vertical differences form a right triangle. The shoelace sum adds signed triangle areas around ordered vertices; repeat the first vertex at the end.

See it: The left side has length $\sqrt{13}$ and slope $3/2$. Shoelace gives $\frac{1}{2}(24 - 12) = 6$.

15 A torus sweeps a circle around an axis



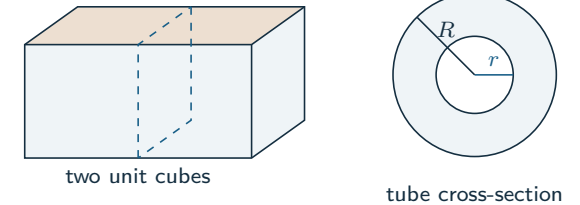
$$V = (\pi r^2)(2\pi R) = 2\pi^2 R r^2$$

$$S = (2\pi r)(2\pi R) = 4\pi^2 R r$$

For a ring torus ($R > r$), multiply the swept disk's area, or its boundary's length, by the distance travelled by its centre. Inner and outer travel distances balance.

See it: R runs from the rotation axis to the tube's centre; r is the tube radius. They are different lengths.

18 Joined faces disappear; hole walls appear



$$V_{\text{joined}} = 2, \quad S_{\text{joined}} = 6 + 6 - 2(1) = 10$$

$$V_{\text{tube}} = \pi(R^2 - r^2)h$$

$$S_{\text{tube}} = 2\pi R h + 2\pi r h + 2\pi(R^2 - r^2)$$

A shared face is counted twice when adding separate surface areas, so subtract twice its area. A through-hole removes volume but adds an inner wall.

See it: The tube has two annular ends, an outer wall, and an inner wall. Here $0 < r < R$.